

Problem 1

Given stoich table find

(1.A) net rates

(1.B) Elementary rate laws if 1, 2 are irreversible and 3 is reversible.

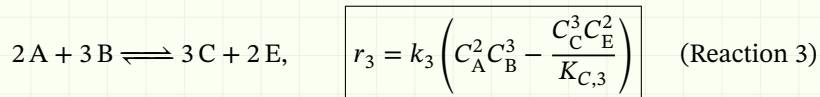
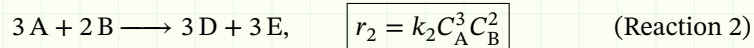
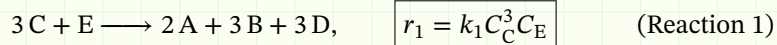
Sp.	ν_{ij}
A	[2, -3, -2]
B	[3, -2, -3]
C	[-3, 0, 3]
D	[3, 3, 0]
E	[-1, 3, 2]

Part A

Net rates can be read from the stoich table as the individual rows:

$$\begin{aligned} r_A &= 2r_1 - 3r_2 - 2r_3 \\ r_B &= 3r_1 - 2r_2 - 3r_3 \\ r_C &= -3r_1 + 0r_2 + 3r_3 \\ r_D &= 3r_1 + 3r_2 + 0r_3 \\ r_E &= -r_1 + 3r_2 + 2r_3 \end{aligned}$$

Part B



Problem 2

Find k at 65°C given the following information:

(2.A) $A = 10^5 \text{ s}^{-1}$, $E_a = 30 \text{ kJ}\cdot\text{mol}^{-1}$.

(2.B) $A = 12\,345 \text{ L}\cdot\text{mol}^{-1}\cdot\text{h}^{-1}$, $E_a = 3 \text{ kcal mol}^{-1}$.

(2.C) $k = 0.14 \text{ min}^{-1}$ at 30°C with $E_a = 2.6 \times 10^4 \text{ J}\cdot\text{mol}^{-1}$.

(2.D) $k = 6.2 \text{ m}^3\cdot\text{kmol}^{-1}\cdot\text{s}^{-1}$ at 85°C with $E_a = 14 \text{ kJ}\cdot\text{mol}^{-1}$.

Part A

Arrhenius form, Eq. (3.9):

$$k = A \exp\left[\frac{-E_a}{RT}\right] = 10^5 \exp\left[\frac{-30000}{(8.314)(338)}\right]$$

$$k = 2.31 \text{ s}^{-1}$$

Note that k has the same units as A .

$$65^\circ\text{C} = 338 \text{ K}$$

Part B

Arrhenius form, Eq. (3.9):

$$k = A \exp\left[\frac{-E_a}{RT}\right] = 12345 \exp\left[\frac{-12552}{(8.314)(338)}\right]$$

$$k = 141.8 \text{ L}\cdot\text{mol}^{-1}\cdot\text{h}^{-1}$$

$$3 \text{ kcal} = 12552 \text{ J}$$

Part C

Use Eq. (3.10) to adjust k given at some temperature T_1 to some other temperature T_2 :

$$k_2 = k_1 \exp\left[\frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2}\right)\right] = 0.14 \exp\left[\frac{2.6 \times 10^4}{8.314} \left(\frac{1}{303} - \frac{1}{338}\right)\right]$$

$$k = 0.408 \text{ min}^{-1}$$

Part D

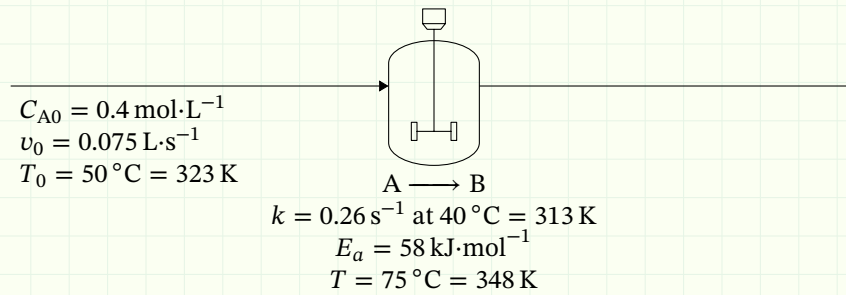
Use Eq. (3.10) to adjust k given at some temperature T_1 to some other temperature T_2 :

$$k_2 = k_1 \exp\left[\frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2}\right)\right] = 6.2 \exp\left[\frac{14000}{8.314} \left(\frac{1}{358} - \frac{1}{338}\right)\right]$$

$$k = 4.69 \text{ m}^3\cdot\text{kmol}^{-1}\cdot\text{s}^{-1}$$

Problem 3

Find CSTR volume given $X_A = 0.62$.



Let's start with a stoichiometric table,

Sp.	ν_i	n_{i0} (mol·s ⁻¹)	n_i (mol·s ⁻¹)
A	-1	0.03	$0.03 - rV$
B	1	0.00	rV
	0	0.03	0.03

$$n_{A0} = C_{A0}v_0 = 0.03 \text{ mol}\cdot\text{s}^{-1}$$

and the rate law,

$$r = kC_A = k\frac{n_A}{v} = 2.45\frac{n_A}{0.075}.$$

$$v = v_0 \text{ for liquids}$$

$$k = 2.45 \text{ s}^{-1} \text{ at } 75^\circ\text{C}$$

We also know from the specification of conversion

$$X_A = 0.62 = 1 - \frac{n_A}{n_{A0}} = 1 - \frac{n_A}{0.03} \Rightarrow n_A = 0.0114 \text{ mol}\cdot\text{s}^{-1}$$

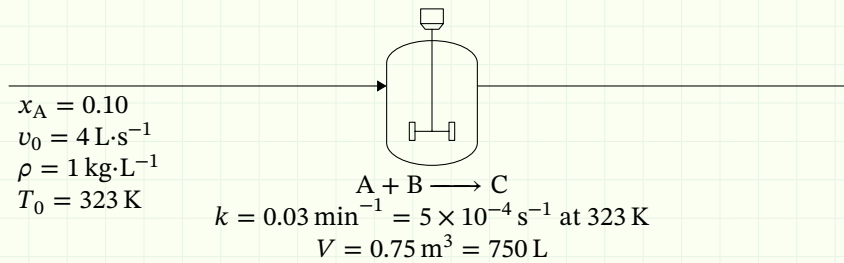
Substituting into the material balance for A yields

$$n_A = n_{A0} + r_A V$$

$$0.0114 = 0.03 - 2.45 \frac{0.0114}{0.075} V$$

from which $V = 0.05 \text{ L}$.

Problem 4



(4.A) Find X_A , n_C .

(4.B) skip

(4.C) Find reactor T , n_C if $E_a = 27 \text{ kJ}\cdot\text{mol}^{-1}$.

Part A

Let's start with a stoichiometric table,

Sp.	ν_i	n_{i0} (mol·s ⁻¹)	n_i (mol·s ⁻¹)
A	-1	19.4	$19.4 - rV$
B	-1	174.8	$174.8 - rV$
C	1	0.0	rV
	-1	194.2	$194.2 - rV$

and a **non-elementary** rate expression,

$$r = kC_A = k \frac{n_A}{v_0}.$$

Substituting into the material balance for A yields

$$n_A = 19.4 - (5 \times 10^{-4}) \frac{n_A}{4} (750) \quad \Rightarrow \quad n_A = 17.8 \text{ mol}\cdot\text{s}^{-1}$$

from which

$$X_A = 1 - \frac{n_A}{n_{A0}} = 1 - \frac{17.8}{19.4} = \boxed{0.08}$$

and

$$n_C = rV = (5 \times 10^{-4}) \frac{17.8}{4} (750) = \boxed{1.7 \text{ mol}\cdot\text{s}^{-1}}$$

You'll get slightly different answers depending on how many digits of precision you carry through the calculation.

Part B

Skip this problem.

Part C

For an adiabatic reactor we must also consider the energy balance according to Eq. (4.2), which for just one reaction reduces to

$$0 = \sum n_{i0} C_{P,i} (T - T_0) + Vr\Delta H$$

There are several ways to get the inlet molar flow rates, all of which will require the molar masses of ethylene oxide (A, 44) and water (B, 18). My approach was to get the total *mass* flow rate m_T as

$$m_T = v_0 \rho = (4)(1) = 4 \text{ kg}\cdot\text{s}^{-1}$$

and the total inlet *molar* flow rate as

$$\begin{aligned}
 n_{T0} &= \frac{m_T}{\text{avg. molar mass}} \\
 &= \frac{4 \times 1000}{0.1(44) + 0.9(18)} \\
 &= 194.2 \text{ mol}\cdot\text{s}^{-1}
 \end{aligned}$$

Be careful with the units: the mass flow had mass units of kg but the molar masses had mass units of grams, hence the factor of 1000 in the numerator.

You could also find n_C from $\xi = n_{A0} - n_A$ and $n_C = \xi$.

adiabatic: $Q = 0$

Since there are only two inlets we can expand the sum to

$$0 = (n_{A0}C_{P,A} + n_{B0}C_{P,B})(T - T_0) + Vr\Delta H$$

We'll need temperature-dependent properties so let's add to the stoichiometric table columns for $\Delta H_{f,i}^\circ$ and C_P from Table D.3 for each species:

Sp.	ν_i	n_{i0} (mol·s ⁻¹)	n_i (mol·s ⁻¹)	$\Delta H_{f,i}^\circ$ (kJ·mol ⁻¹)	$C_{P,i}$ (J·mol ⁻¹ ·K ⁻¹)
A	-1	19.4	$19.4 - rV$	-52.6	48.0
B	-1	174.8	$174.8 - rV$	-285.8	75.0
C	1	0.0	rV	-454.8	150.0
	-1	194.2	$194.2 - rV$	-116.4	27.0

In the bottom row are also calculated ΔH° and ΔC_P . The material balance on A worked well before but now has two unknowns, n_A and T . The energy balance will have the same two unknowns, so that'll constitute a system of two independent equations with two unknowns:

$$n_A = 19.4 - k \frac{n_A}{4} (750)$$

$$0 = (19.4(48) + 174.8(75))(T - 323) + k \frac{n_A}{4} (750)\Delta H$$

where

$$k = (5 \times 10^{-4}) \exp \left[\frac{27000}{8.314} \left(\frac{1}{323} - \frac{1}{T} \right) \right]$$

$$\Delta H = -116400 + 27(T - 298)$$

It's ok to solve this problem in MATLAB but it can struggle to converge (see note below). Let's take our algebra one step further and rearrange the material balance to get n_A as a function of only T :

$$n_A = \frac{19.4}{1 + \frac{750k}{4}}$$

When substituted into the energy balance we've got one equation with just one unknown, T . We can plot the energy balance in rootfinding form to estimate the solution temperature, then use `fsolve` to find that temperature. From the plot (see sidebar) the temperature looks like it's at about 350 K so we use that as the initial guess.

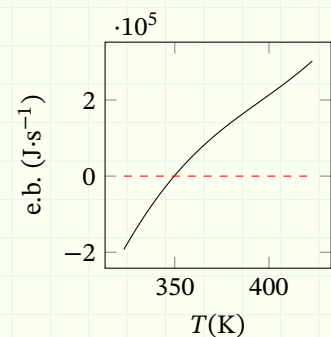
```

1 T = linspace(323,423); % K
2 eb = rff(T); % J/s
3 plot(T, eb);
4 yline(0, 'r--'); % suggests solution is around 350 K
5
6 T = fsolve(@rff, 350);
7
8 function eb = rff(T)
9     k = (5e-4)*exp(27000/8.314*(1/323-1./T)); % 1/s
10    nA = 19.4./(1+750*k/4); % mol/s
11    dH = -116400 + 27*(T-298); % J/mol
12    eb = (19.4*48+174.8*75)*(T-323) + k.*nA/4*750.*dH; % J/s
13 end

```

This code produces $T = 350 \text{ K} = 77^\circ\text{C}$, and substituting back into the formula for n_A yields $n_A = 16.1 \text{ mol}\cdot\text{s}^{-1}$, from which $n_C = \xi = n_{A0} - n_A = 3.3 \text{ mol}\cdot\text{s}^{-1}$.

Use Eq. (4.22) to adjust ΔH for temperature, and ensure units are $\text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$ to match C_P .



Comment on (4.C)

If we wanted to solve (4.C) as a system of two equations and two unknowns we may bump into an issue called *scaling*. It's described in §4.6 in your book and the basic problem is that when guesses are substituted into the system of equations the material balance gives numbers around 10 but the energy balance gives numbers around 100,000.

The `fsolve` function struggles to find a solution because no matter which guesses it picks it concludes the guess is horrible because the energy balance is nowhere near 0, but the material balance is *always* near zero (relatively speaking). It needs both equations to give numbers of approximately the same order of magnitude, or put another way: it needs the problem to be *scaled* appropriately.

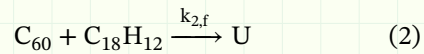
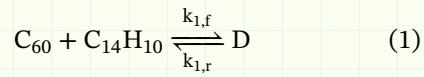
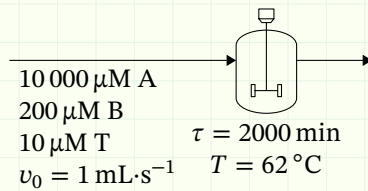
To fix a scaling issue you first have to recognize it. One way to do that is to try a few guesses in your local function all by yourself and see if the output provides numbers of roughly the same order of magnitude. To actually correct it once discovered isn't too bad because you can change units of one of the equations until things work out. For example, the code below changes the units of the energy balance to $\text{kJ}\cdot\text{mol}^{-1}$ instead of $\text{J}\cdot\text{mol}^{-1}$ and everything works just fine.

```

1 Y0 = [17; 333]; % [nA; T] guesses
2 Ysol = fsolve(@rff, Y0);
3 fprintf('nA = %.1f mol/s, T = %.1f K\n', Ysol(1), Ysol(2));
4
5 function sysOfEq = rff(Y)
6     nA = Y(1); % mol/s
7     T = Y(2); % K
8
9     k = 5e-4*exp(27000/8.314*(1/323-1./T)); % 1/s
10    dH = -116400+27*(T-298); % J/mol
11
12    eq1 = 19.4-k*nA/4*750 - nA; % mol/s
13    eq2 = (19.4*48+174.8*75)*(T-323) + k*nA/4*750*dH; % J/s
14    sysOfEq = [eq1; eq2/1000]; % scale eb to kJ/s to help
15    fsolve
16 end

```

Problem 5



Rate constant	A	E_a
$k_{1,f}$	$6.310 \text{ L} \cdot \mu\text{mol}^{-1} \cdot \text{s}^{-1}$	$59 \text{ kJ} \cdot \text{mol}^{-1}$
$k_{1,r}$	$7.943 \times 10^{13} \text{ s}^{-1}$	$140 \text{ kJ} \cdot \text{mol}^{-1}$
$k_{2,f}$	$25.12 \text{ L} \cdot \mu\text{mol}^{-1} \cdot \text{s}^{-1}$	$49 \text{ kJ} \cdot \text{mol}^{-1}$

(5.A) V , volume of fluid entering ever second, compare to everyday object.

(5.B) Stoich table

(5.C) Outlet concentrations of everything but toluene

Part A

The reactor volume is given by Eq. (4.5) as

$$V = \tau v_0 = (2000 \text{ min}) (1 \text{ mL} \cdot \text{s}^{-1}) = \boxed{120 \text{ L}}$$

WolframAlpha suggests this is a little smaller than a typical bathtub. In one second the volume entering the reactor is $\boxed{1 \text{ mL}}$, which is smaller than a soy sauce packet.

The purpose of this problem was to highlight the rather surprising disconnect in sizes here: about every few seconds a soy sauce packet is added to a bathtub, and a soy sauce packet is removed from the bathtub. That's not a very big flow!

Part B

Stoichiometric table for two reactions:

species	abbr.	ν_{ij}		n_{i0} ($\mu\text{mol} \cdot \text{s}^{-1}$)	n_i ($\mu\text{mol} \cdot \text{s}^{-1}$)
buckyballs	B	-1	-1	0.20	$0.20 - (r_1 + r_2)V$
anthracene	A	-1	0	10.00	$10 - r_1V$
tetracene	T	0	-1	0.01	$0.01 - r_2V$
desired	D	1	0	0.00	r_1V
undesired	U	0	1	0.00	r_2V
		-1	-1	10.21	$10.21 - (r_1 + r_2)V$

Each molar flow rate is calculated as $C_i v_0$.

Part C

The two elementary rate laws are

$$\begin{aligned} r_1 &= k_{1f} C_B C_A - k_{1r} C_D \\ r_2 &= k_{2f} C_B C_T \end{aligned}$$

There's not really much hope of solving this by hand so we'll just ask `fsolve` to solve a system of five equations (*i.e.*, five material balances) for five unknowns (*i.e.*, the five molar flow rates). We can then calculate outlet concentrations when we're done. Pressing right along:

```

1 nGuess = [0.2; 10; 0.01; 0; 0]; % umol/s [B A T D U]
2 n = fsolve(@rff, nGuess)
3
4 function out = rff(n)
5     R = 8.314; % J/mol-K
6     T = 62+273; % K
7     v0 = 1e-3; % L/s
8     V = 120; % L
9     C = n/v0; % umol/L
10    k1f = 6.310 * exp(-59e3/(R*T)); % L/umol-s
11    k1r = 7.943e13 * exp(-140e3/(R*T)); % 1/s
12    k2f = 25.12 * exp(-49e3/(R*T)); % L/umol-s
13    r1 = k1f*C(1)*C(2) - k1r*C(4); % umol/L-s
14    r2 = k2f*C(1)*C(3); % umol/L-s
15    out = [ 0.2 + (-r1-r2)*V - n(1); % C60 (B)
16           10 + (-r1)*V - n(2); % anthracene (A)
17           0.01 + (-r2)*V - n(3); % tetracene (T)
18           (r1)*V - n(4); % desired (D)
19           (r2)*V - n(5)]; % undesired (U)
20 end

```

This produces the outlet molar flow rates shown in the table below, from which the corresponding concentrations are calculated as $C_i = n_i/v_0$.

species	abbr.	ν_{ij}		n_{i0} ($\mu\text{mol}\cdot\text{s}^{-1}$)	n_i ($\mu\text{mol}\cdot\text{s}^{-1}$)	C_i (μM)
buckyballs	B	-1	-1	0.20	0.034	34
anthracene	A	-1	0	10.00	9.841	9841
tetracene	T	0	-1	0.01	0.003	3
desired	D	1	0	0.00	0.159	159
undesired	U	0	1	0.00	0.007	7

Remember $v = v_0$ for liquids only.